Statistical Analysis of Datasets

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Motivation: How to make stronger claims about the data

- How to represent data: the expressive power of different representations
- Social networks: advanced notions on community detection
- Entity matching between different datasets
- Natural language processing to extract and classify information from text sources
- **An introduction to statistics in social research:** whether the data provides sufficient evidence to reject a particular hypothesis (claim). Online books: [OpenStax, 2024, OpenIntro, 2024, Ani Adhikari, John DeNero, and David Wagner, 2024]

Simple descriptive statistics

Simple descriptive statistics

Given in input a **series of data points**, we can compute:

- **mean**: the average of all numbers in the series
- **median**: we sort the data points from smallest to largest. If the number of elements odd, the median is the middle data point. If the number of data points is even, the median is the average of the two middle data points.
- **min, max, frequent elements** etc. The most frequent element is also called the mode.
- **quartiles and interquartile range**

Quartiles and interquartile range

How to better describe a series of data points?
To calculate the quartiles:

1. sort the data points in increasing order and compute the median M (called also the second quartile Q2)
2. the first quartile Q1 is the median of the data points located to the left of M in the sorted list
3. the third quartile Q3 is the median of the data points located to the right of M in the sorted list
4. the interquartile range (IQR) is defined as: IQR = Q3 – Q1

Five-Number Summary

The **maximum and minimum** values alone do not give enough information about the distribution as a whole.
Likewise, the **median and the quartiles** do not give information about the tails of the distribution.

The **five-number summary** combines all of these values. It includes the minimum, the first quartile (Q1), the median (Q2), the third quartile (Q3), and the maximum (Q4).

Interesting distributions of data points in the real world

The **normal distribution** describes the distribution of values for many natural phenomena in a wide range of areas, including biology, physical science, mathematics, finance and economics, such as:

- biology, the logarithm of various variables such as measures of size of living tissue (length, height, skin area, weight)
- IQ scores, etc.

Given a series of data points, there are tests for normality.

The normal distribution

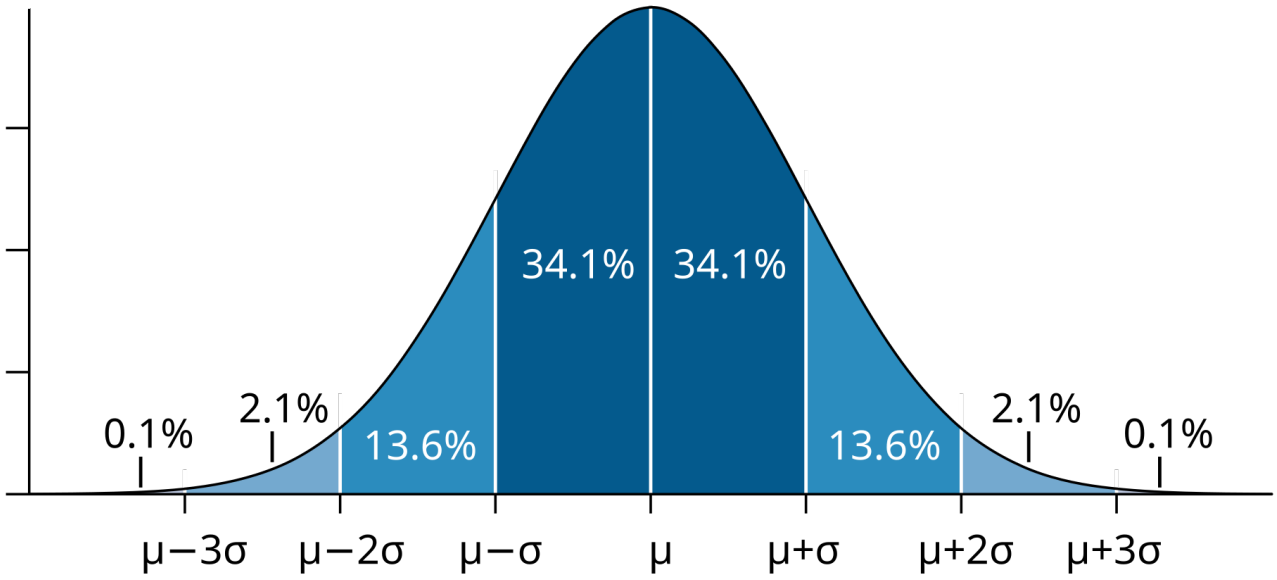


Figure: Probability density of normal distribution: probability of the random variable falling within a particular range of values

A **normal distribution** is a continuous probability distribution in which most data points cluster toward the middle of the range. It is described by its mean and standard deviation.

Standard deviation

Standard deviation: is a measure of variability in a distribution. In the normal distribution, it defines the width of the distribution.

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

where

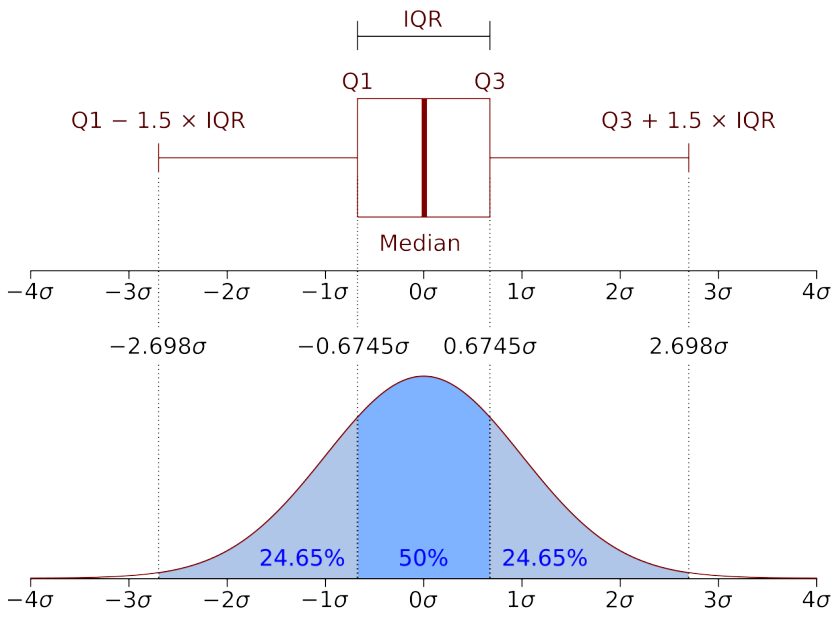
$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Variance is the square of the deviation.

Outliers in normal distribution

An **outlier** is a data point that is significantly different from the other data points. Informal definition: inspect the data points.

Detection: via the standard deviation (outside $\pm 3\sigma$), or the interquartile range (outside ± 1.5 IQR).



Making statistical claims about the data

Hypothesis testing

We split the students into two groups: students who sit on the left of the class and students who sit on the right.

If p_{left} is the proportion of students who love ice cream and sit on the left side of the classroom and p_{right} is the proportion of students who love ice cream and sit on the right side of the classroom, should $p_{right} = p_{left}$?

No, we would probably observe a small difference due to chance.

When is a difference between two groups significant? E.g. how can we say if the gender/race/political orientation of a person affected how this person was treated?

Hypothesis testing

A hypothesis test is a statistical technique used to evaluate competing claims using data.

The null hypothesis H_0 often represents either a skeptical perspective or a claim of "no difference" to be tested. E.g. gender/race/political orientation did not affect the treatment a person received (e.g. how often people with this attribute receive salary increases).

The alternative hypothesis H_A represents an alternative claim under consideration and is often represented by a range of possible values for the value of interest.

Comparing means of samples

Student's t-test (also known as t-test)

Student's t-test (also known as t-test): is a **statistical test for testing for significant differences between the means of two groups** (which we suppose follow a normal distribution). The t-test uses the t-distribution, which is generalizes the normal distribution and has mean 0.

Figure: Source:`https://datatab.net/tutorial/t-test`

One-Sample t-Test

To compare the mean of our sample with a known mean μ_0, we compute the the t-test formula:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

where:

- $\bar{x}$ = sample mean
- μ_0 = population mean
- s = sample standard deviation
- n = sample size

The meaning of t

When you run a t-test, the formula calculates a t-statistic (or t-value). We can see t-value as a signal-to-noise ratio:

- The Signal (Numerator): The difference between your group averages.
- The Noise (Denominator): The variability (standard error) within your groups.

A t-value close to 0: There is very little difference between your groups. The "signal" is weak compared to the "noise."

Outliers r-value: values that are very far away from 0, values that are outside the 95% interval (i.e. the interval in which 95% of data lies). We say that these values have a small probability of occurring, i.e. their p-value is less that 0.05.

Two-Sample t-Test (Independent Samples)

For two independent samples, the t-test formula is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where:

- $\bar{x}_1, \bar{x}_2$ are the sample means
- s_1^2, s_2^2 are the sample variances
- n_1, n_2 are the sample sizes

Two-Sample t-Test (Dependent Samples)

For two dependent samples, the t-test formula is:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

where:

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i$$

$$s_d = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2}$$

- d_i = difference between each paired sample
- $\bar{d}$ = mean of the differences between paired observations.
- s_d = standard deviation of the differences.
- n = number of paired observations.

Hypothesis testing for the t-test

Hypotheses:

$$H_0 : \bar{x}_1 = \bar{x}_2$$

$$H_A : \bar{x}_1 \neq \bar{x}_2$$

Rejection Rule for H_0:

$$p < \alpha$$

Where the **p-value is the probability of obtaining a test statistic t at least as extreme as the observed value**, assuming the null hypothesis is true. The p-value is obtained from precomputed t-distribution table.

If p is small e.g. $p < 0.01$ or $p < 0.05$, the probability is low that we observe something similar given the null hypothesis.

Two-Sample t-Test (Independent Samples)

We want to test whether there is a significant difference between the average exam scores of two groups of students (Group 1 and Group 2) after they were taught using different methods.

- **Group 1 (Traditional Teaching):**
  - Sample size (n_1): 10 students
  - Sample mean ($\bar{x}_1$): 75
  - Sample standard deviation (s_1): 8
- **Group 2 (New Teaching Method):**
  - Sample size (n_2): 12 students
  - Sample mean ($\bar{x}_2$): 80
  - Sample standard deviation (s_2): 7

The formula for the t-statistic is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \approx -1.55$$

For this t value, $p \approx 0.14$

ANOVA

The ANOVA (Analysis of Variance) test: **differences between the means of three or more groups**.

Is at least one group mean significantly different from the others?

ANOVA is commonly used when comparing multiple groups in experiments and it is an extension of the t-test for more than two groups.

ANOVA

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

where $\mu_1, \mu_2, \ldots, \mu_k$ are the means of the k groups.

$$H_A : \text{At least one group mean is different}$$

F-statistic: is the ratio of the between group variance to the within group variance (follows the F-distribution).

$$F = \frac{\text{Between group variance}}{\text{Within group variance}}$$

A large F-statistic indicates that the between group variability is large relative to the within group variability, suggesting a significant difference between the group means.

ANOVA

- **Between-group variance** (SS_{between}):

$$SS_{\text{between}} = \sum_{i=1}^{k} n_i(\bar{X}_i - \bar{X})^2$$

  where n_i is the number of observations in the i-th group, $\bar{X}_i$ is the mean of the i-th group, and $\bar{X}$ is the overall mean .
- **Within-group variance** (SS_{within}):

$$SS_{\text{within}} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

  where X_{ij} is the j-th observation in the i-th group and $\bar{X}_i$ is the mean of the i-th group.
- **Determine the p-value** using the F-distribution.
- **Compare** the p-value to the significance level ($e.g. \alpha = 0.05$).

t-test and ANOVA assumptions

t-test: It can be used under the conditions that the sample size is large (or the observations are normally distributed), and if the two samples have similar variances.

Welch's t-test is essentially the same thing as Student's t-test except that it is used when the two samples do not have similar variances.

Mann-Whitney test: should only be used when all assumptions for all previous tests are violated: small sample size/not normally distributed.

ANOVA: same assumptions as t-test, when they do not hold we use the **Kruskal–Wallis test**.

Shapiro-Wilk Test for Normality

The **Shapiro-Wilk test** is used for checking whether a dataset is **normally distributed**.

- H_0: The data follows a **normal distribution**.
- H_A: The data **does not follow a normal distribution**.

The test produces a test statistic, denoted as W, which measures how well the data matches a normal distribution.

Shapiro-Wilk Test for Normality

The Shapiro-Wilk test statistic W is calculated as:

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_i\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Where:

- x_i are the **ordered sample values** (sorted in ascending order).
- $\bar{x}$ is the **sample mean**.
- a_i are the **weights** derived from the normal distribution.

The numerator represents the **weighted sum** of the sample data, and the denominator represents the **total variation** in the data.

Interpret the p-value: Based on the computed W statistic, the test will calculate a **p-value**.

Comparing frequencies of events

Fisher's exact test

Contingency table is a type of table in a matrix format that displays the multivariate frequency distribution of the variables.

	Group 1	Group 2
Category 1	a	b
Category 2	c	d

	Smoker	Non-Smoker	Total
Male	8	2	10
Female	1	9	10
Total	9	11	20

Fisher's Exact Test: is there is a significant association between two categorical variables in a contingency table.

Fisher's exact test

The hypergeometric distribution is used to calculate probabilities when you are drawing from a finite population without replacement, e.g. Fisher's Exact Test.

In a 2x2 contingency table, a fixed number of items need to be assigned to cells (categories), and we want the probability of a specific distribution of these items.

Could the observed distribution occur by chance, assuming the null hypothesis that the variables are independent?

Fisher's exact test

	Group 1	Group 2
Category 1	a	b
Category 2	c	d

$$p = \frac{\binom{a+b}{a}\binom{c+d}{c}}{\binom{a+b+c+d}{a+c}}$$

- $\binom{a+b}{a}$ is the number of ways to choose a "successes" (e.g., Male Smokers) from the total $a+b$ (Category 1).
- $\binom{c+d}{c}$ is the number of ways to choose c "successes" (e.g., Female Smokers) from the total $c+d$ (Category 2).
- $\binom{a+b+c+d}{a+c}$ is the total number of ways to select $a+c$ "successes" from the total population of $a+b+c+d$.

Fisher's exact test

Hypotheses

- **Null Hypothesis (H_0):** There is no association between the two variables. The variables are independent.
- **Alternative Hypothesis (H_A):** There is an association between the two variables. The variables are dependent.

Fisher's Exact Test Formula

$$p = \frac{\binom{a+b}{a}\binom{c+d}{c}}{\binom{a+b+c+d}{a+c}}$$

Where a, b, c, and d are the observed counts in the 2x2 table:

Decision Rule

- Calculate the p-value for the observed table: the sum of the probabilities for all tables (with the same sum of values of lines and columns) having a probability equal to or smaller than p.

Correlation of samples

Correlation

Correlation or dependence: statistical relationship, whether causal or not, between two random variables. If the variables are independent, there is no correlation.
Correlation may indicate any type of association, however, usually refers to the degree to which a pair of variables are linearly related.

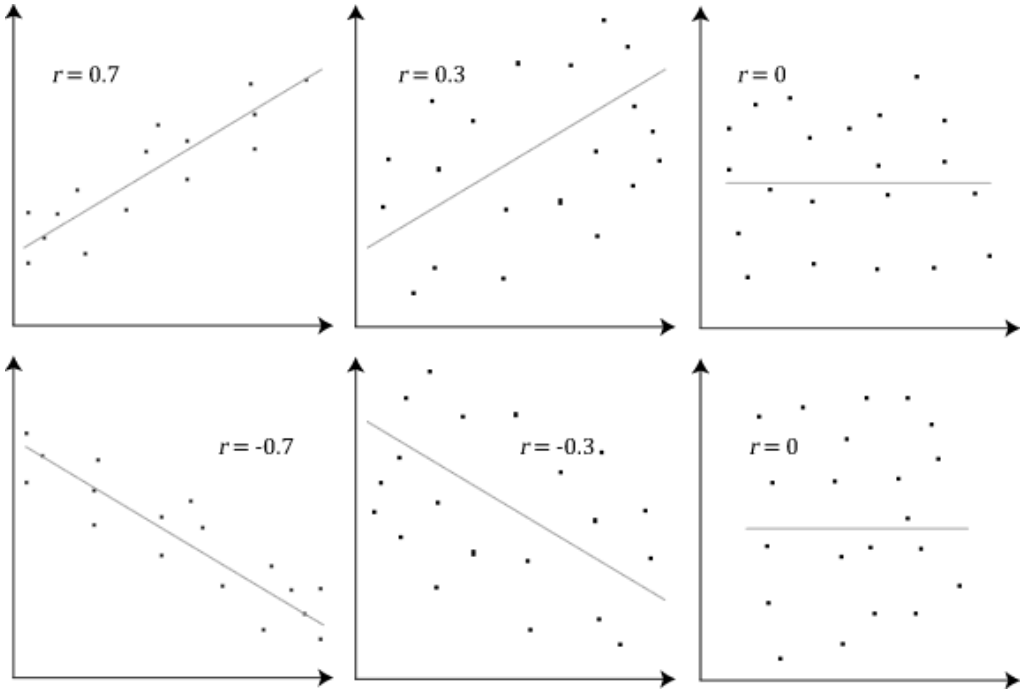


Figure: Different correlation coefficients

Correlation

Correlation ranges from -1 to 1.
A value of -1 indicates a perfect negative correlation. A value of 0 indicates no correlation. A value of 1 indicates a perfect positive correlation.

Correlation does not imply **causation**, however it can identify a potential relation between variables.

Example: A researcher investigates the relationship between the amount of exercise people do weakly and their overall physical fitness. They could use a correlation coefficient to determine whether there is a relation.

Pearson correlation coefficient

Pearson correlation coefficient (also called **Pearson's** r) is a statistical measure that quantifies the strength and direction of the linear relationship between two continuous variables.

The Pearson correlation coefficient is given by:

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2}\sqrt{\sum(Y_i - \bar{Y})^2}}$$

where:

- X_i and Y_i are individual data points for the two variables,
- $\bar{X}$ and $\bar{Y}$ are the means of X and Y,
- The numerator represents the **covariance** between X and Y, measuring how they vary together,
- The denominator normalizes the covariance using the **standard deviations** of X and Y.

Pearson correlation coefficient

The correlation coefficient r takes values between -1 and 1:

- $r = 1$: Perfect positive correlation (as X increases, Y increases proportionally).
- $r = -1$: Perfect negative correlation (as X increases, Y decreases proportionally).
- $r = 0$: No linear correlation (no predictable relationship between X and Y).

Intermediate values indicate varying degrees of correlation:

- $0.7 \leq r < 1$: Strong positive correlation.
- $0.4 \leq r < 0.7$: Moderate positive correlation.
- $0 \leq r < 0.4$: Weak positive correlation.
- $-0.4 < r \leq 0$: Weak negative correlation.
- $-0.7 < r \leq -0.4$: Moderate negative correlation.
- $-1 < r \leq -0.7$: Strong negative correlation.

p-value for Pearson Correlation

The **Pearson correlation coefficient** (r) has an associated **p-value**, which tests the null hypothesis that there is no linear relationship between the two variables.
For significance of the correlation, we use the **t-statistic**:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

where:

- r is the Pearson correlation coefficient,
- n is the sample size,
- t follows a **Student's t-distribution**.

p-value is the probability of obtaining a test statistic t at least as extreme as the observed value, assuming the null hypothesis is true.

Correlation does not imply causation

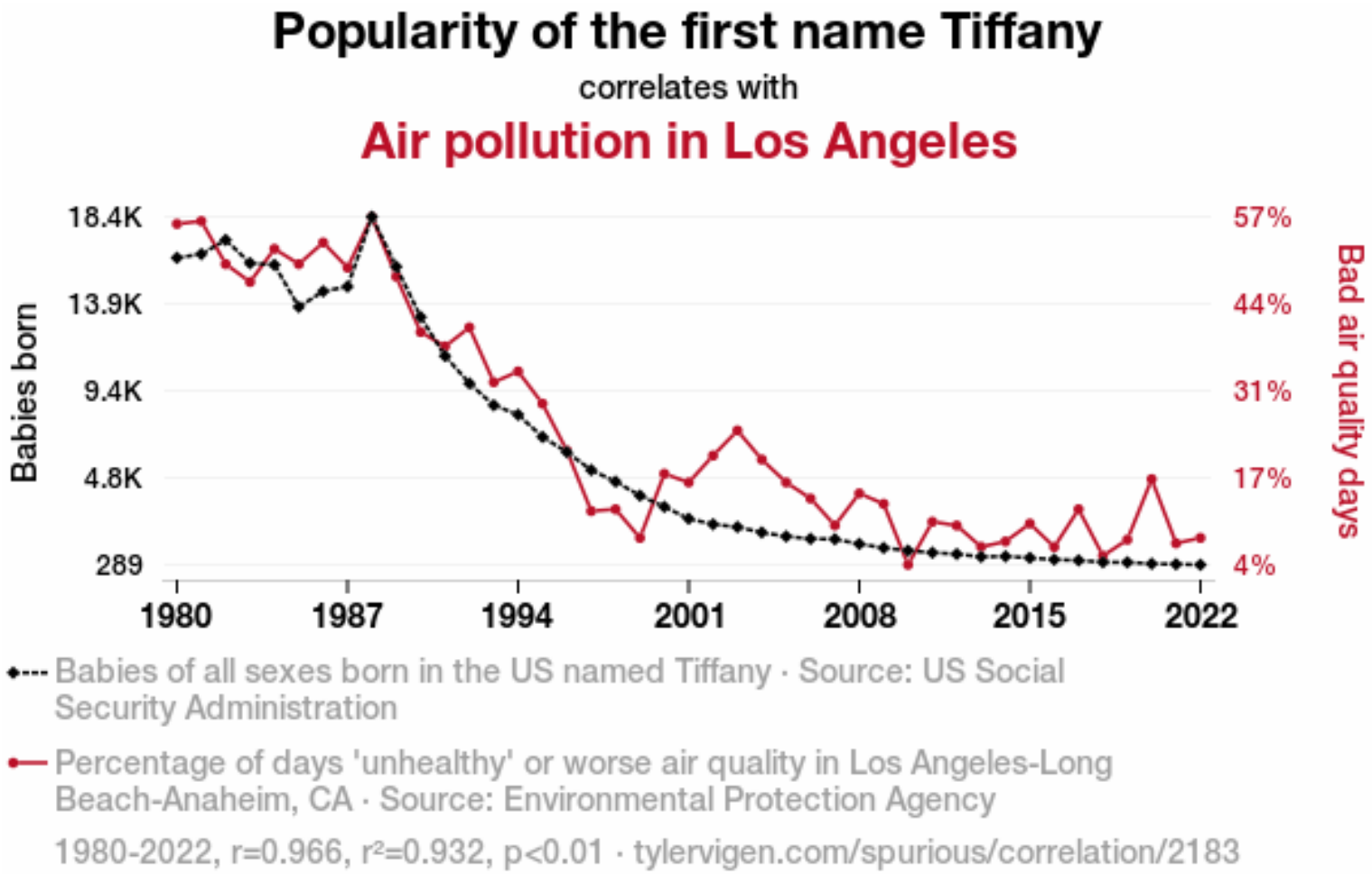


Figure: Source: `https://www.tylervigen.com/spurious-correlations`

Correlation does not imply causation

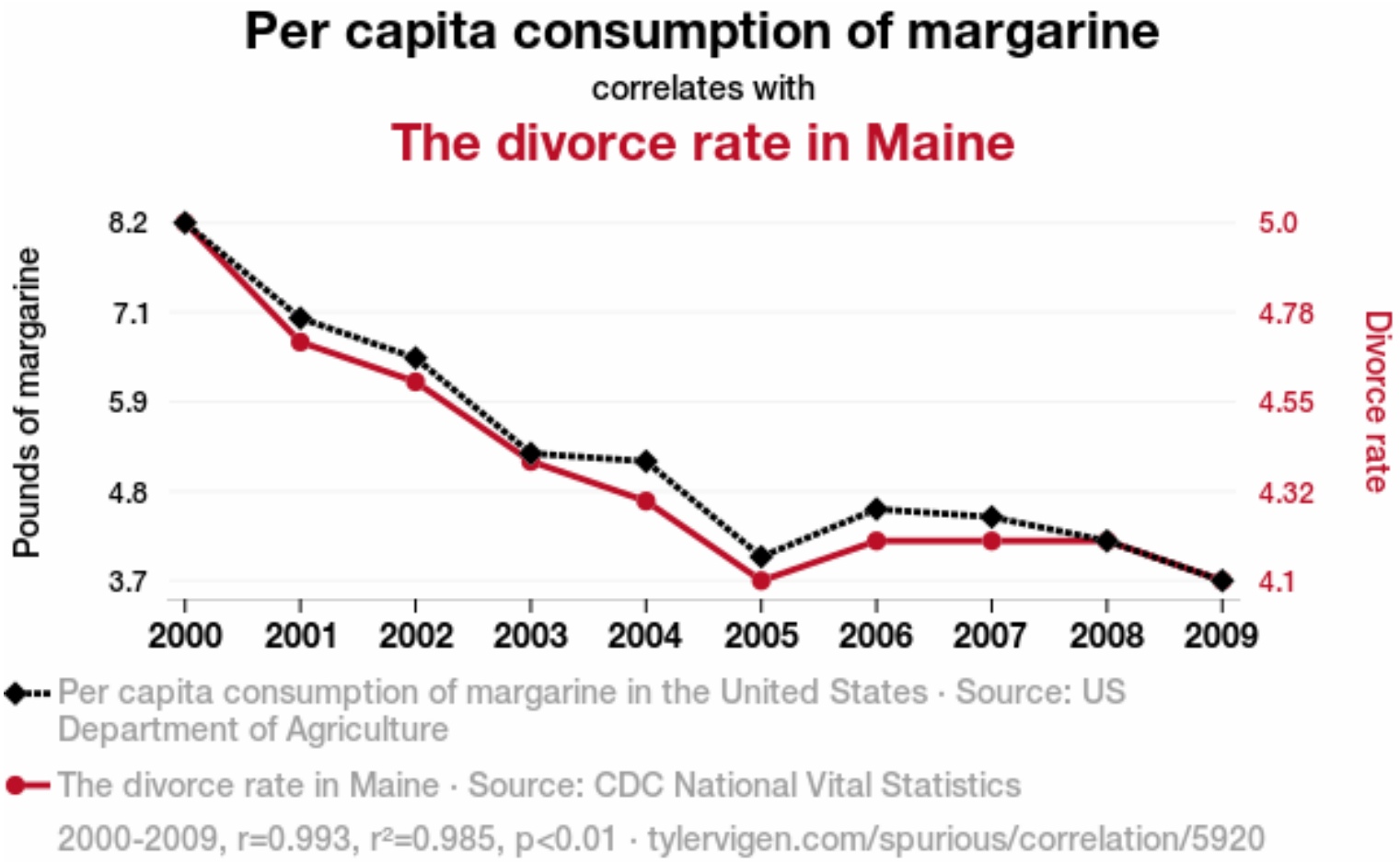


Figure: Source: `https://www.tylervigen.com/spurious-correlations`

Granger causality

The Granger causality test: determining whether one time series is useful in forecasting another. Forecasting is not as strong as causation.

A variable X is said to **Granger-cause/forecasts** another variable Y if past values of X improve the prediction of Y, beyond what is possible using only past values of Y itself.

Granger causality is tested by comparing two models.

Granger causality

Restricted Model

$$Y_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i Y_{t-i} + \epsilon_t$$

where:

- Y_t is the dependent variable at time t,
- Y_{t-i} are its past values (lags),
- ϵ_t is the error term.

Unrestricted Model

$$Y_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i Y_{t-i} + \sum_{j=1}^{q} \beta_j X_{t-j} + \epsilon_t$$

where:

- The additional terms X_{t-j} (past values of X) are included.

Granger causality

If adding lagged values of X significantly improves the prediction of Y, then X is said to **Granger-cause/forecast** Y.

- H_0: X does not Granger-cause Y, meaning that past values of X do not improve the prediction of Y.
- H_A: X Granger-causes Y, meaning that past values of X help in predicting Y.

An statistical test is used to check if the coefficients β_j are significantly different from zero.
A small p-value of the test (e.g., $p < 0.05$) indicates that we reject H_0, meaning X Granger-causes Y.

Project

Class Overview

- How to represent data: the expressive power of different representations
- Social networks: advanced notions on community detection
- Entity matching between different datasets
- Natural language processing to extract and classify information from text sources
- An introduction to statistics in social research

What data can we use?

Open Government Data (OGD) *is a philosophy - and increasingly a set of policies - that promotes transparency, accountability and value creation by making government data available to all. Public bodies produce and commission huge quantities of data and information. By making their datasets available, public institutions become more transparent and accountable to citizens.*

`https://www.data.gouv.fr/fr/`
`https://data.europa.eu/en`
`https://data.worldbank.org/`
`https://www.who.int/data/gho/`

You can of course find data on other sites!

Project

Assessment Through the **project** to be carried out in pairs, using the tools introduced, and based on open data. The subject of the project is chosen by the pair.

- Integration of multiple data sources
- Linking entities across the data sources
- In depth analysis of the data (also known as graph/data mining)
- Hypothesis testing using statistical methods
- Cool visualizations

At the end you will submit a project report and you will have to do a presentation. Both will be graded.

References I

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